

Documentation: ARIR/BRIR measurements for 2.5D wave field synthesis

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This dataset includes 120 directional B-format RIRs, i.e. third-order and fifth-order Ambisonic room impulse responses respectively (third and fifth-order ARIRs) and 120 directional binaural RIRs, measured at 5 receiver positions and 24 source positions each, c.f. Figure 1.

The measurements were carried out at the IEM lecture hall, a $6.5\text{ m} \times 7.6\text{ m} \times 2.5\text{ m}$ room with reverberation time $T_{60} \approx 0.5\text{ s}$ of the Institute of Electronic Music and Acoustics (IEM) in Graz, Austria.

24 Genelec 8020 loudspeaker, set up as linear array for 2.5D wave field synthesis, were used as sound sources. The measurements were done with a Zoom H3-VR and a Neumann KU100 on a line parallel to the array in a distance of 2 m for 5 position at intervals of 0.9 m. Since the ZOOM-H3VR microphone offers a first-order B-format output, a resolution enhancement was done by the 2DSE2 algorithm (<https://phaidra.kug.ac.at/o:104572>). The ARIRs and BRIRs were measured with exponential sweeps with a duration of $T = 1\text{ s}$ between 20 Hz and 22 Hz.

In the database the sound sources are numbered from right to left, i.e. loudspeaker 1 is right, loudspeaker 24 is left, and the positions are numbered from left to right.

Each folder also consist a `.conf`-file, which can be used for convolution with Matthias Kronlachner's Plugin `mcfx_convolver` from his mcfx Plguin Suite (<http://www.matthiaskronlachner.com/?p=1910>) inside a DAW like REAPER.

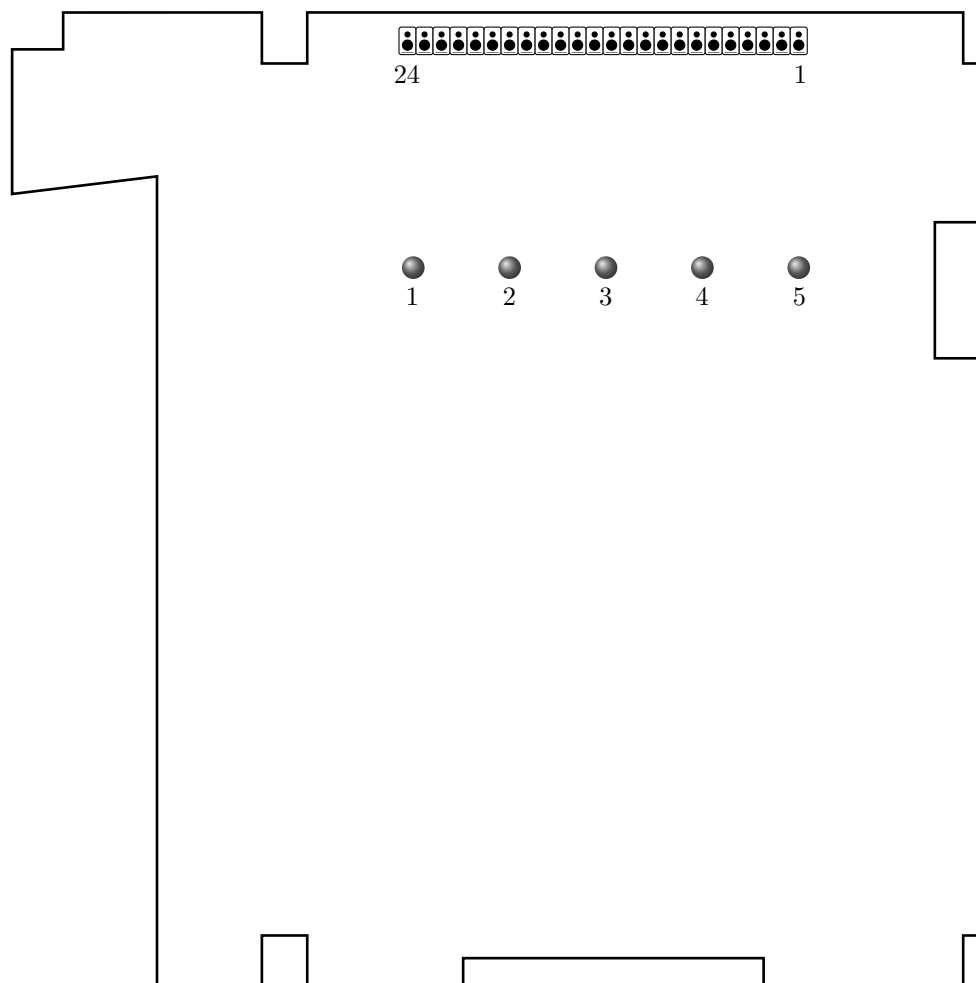


Figure 1: Loudspeaker arrangement for ARIR/BRIR measurement of a linear array for 2.5D wave field synthesis at IEM lecture hall



Figure 2: ARIR/BRIR measurement at IEM lecture hall